# Satya Venkata Raja Maneesh Pyla

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## Research Summary

Ph.D. candidate in theoretical chemistry studying how low-energy electrons attach to molecules and cause damage. I use single- and multireference wave-function methods to characterize electronic resonances (their energies, widths and decay channels) in heterocycles and solvated biomolecules. My work has produced four peer-reviewed papers (two first-author) and two journal covers. I am currently extending it toward machine-learning-accelerated resonance methods, core-excited resonances and non-adiabatic dynamics.

## Education

**Temple University** *Philadelphia, USA*
Ph.D. in Chemistry (Advisor: Prof. Spiridoula Matsika) *2022 – Expected 2027*
**Dissertation focus:** Characterization of electronic resonances using single and multireference wave-function methods.

**Indian Institute of Science Education and Research (IISER) Tirupati** *Tirupati, India*
BS–MS in Chemistry (Advisor: Dr. Raghunathan Ramabhadran) *2017 – 2022*
**Master's thesis:** Density Functional Theory (DFT) investigation of water-in-bisalt (WIBS) electrolyte systems.

## Publications

**5. Maneesh Pyla** and Spiridoula Matsika. *"Can ML based Gaussian Process Regression methods accelerate Stabilization Approaches."* ***in preparation*** **(2026).**

**4.** Kanishka Kuda Vidanelay, **Maneesh Pyla**, Cate Sara Anstöter, and Spiridoula Matsika. *"Revisiting the Nature of Low-Lying π Resonances in Microsolvated Uracil."* ***J. Phys. Chem. Lett.*** **17 (15): 4561–4568 (2026)***(**cover**)*.

**3. Maneesh Pyla** and Spiridoula Matsika. *"Partial Widths of Shape Resonances in Pyridine and Uracil Using the Stabilization Method."* ***J. Chem. Phys.*** **164, 014308 (2026).**

**2.** Divya Tripathi, **Maneesh Pyla**, Achintya Kumar Dutta, and Spiridoula Matsika. *"Impact of solvation on the electronic resonances in uracil."* ***Phys. Chem. Chem. Phys.,*** **27, 3588–3601 (2025).** *(Journal back cover)*.

**1. Maneesh Pyla** and Spiridoula Matsika. *"Exploring electronic resonances in pyridine: Insights from orbital stabilization techniques."* ***J. Chem. Phys.*** **161, 154306 (2024).**

## Awards, Fellowships, & Grants

**Telluride School on Theoretical Chemistry 2025**. Highly selective summer school in theoretical chemistry; selected as one of 30 participants from a competitive applicant pool.

**PROM Funding (MolEx 2025)**. Fellowship to attend MolEx workshop (in Poland) on theoretical modeling of excited states and related computational tools.

**Erasmus Mundus Joint Master Degree (MESC+ 2022)**. Admitted to the international program in Materials for Energy Storage and Conversion involving multiple partner universities and research centers across Europe and beyond. (offer declined)

## Presentations

**Gordon Research Conference: Molecular Interactions and Dynamics**.
Exploring Electronic Resonances in Organic Molecules Using the Stabilization Method (poster)

**Penn Conference in Theoretical Chemistry (PCTC)**.
Exploring Electronic Resonances in Organic Molecules Using the Stabilization Method (poster)

**64th Sanibel Symposium 2025**.
Talk: Exploring Electronic Resonances in Pyridine: Insights from Orbital Stabilization Techniques. (**Selected Hot-Topic Talk**)

**Molecular Excited States Workshop (MolEx 2025)**, Toruń, Poland.
Exploring Electronic Resonances in Pyridine: Insights from Orbital Stabilization Techniques (poster)

## Technical Skills

- **Electronic Structure Methods:** DFT, MP2, EOM-EA-CCSD, RASSCF, CASSCF, XMS-CASPT2
- **Resonance Analysis:** Stabilization Methods, Complex Analysis
- **Non-adiabatic Dynamics:** Trajectory Surface Hopping
- **Software:** Q-Chem, OpenMolcas, Molpro, Gaussian, Terachem, Newton-X, Columbus, ORCA
- **Programming:** Python, Bash, and Matlab
- **High-Performance Computing:** Linux, SLURM/PBS schedulers
- **Machine Learning:** GPRs using scikit-learn

## Workshops

**Deep Modeling for Molecular Simulation 2023**

Department of Chemistry, Princeton University

Hands-on experience with Deep-MD simulations and electronic-structure workflows using DeepMD and Quantum ESPRESSO.

**Photo-Induced Molecular Dynamics: Experiment, Theory, and Computational Modeling 2025**

Rutgers University, Newark (Organized by Prof. Neepa Maitra)

Training in photo-induced molecular dynamics and coherence effects in excited-state processes.

**Molecular Excited States (MolEx 2025) Workshop**

Nicolaus Copernicus University, Poland

Hands-on experience with ORCA, Newton-X, CobraMM, and MLatom for excited-state and nonadiabatic dynamics simulations.

## Schools and Symposiums

**64th Sanibel Symposium 2025**

University of Florida

- Exposure to current developments in electronic structure theory, quantum computing, and machine-learning approaches.

**Virtual Winter School on Computational Chemistry 2025**

University of Edinburgh

- Attended lectures focused on electronic resonances and related computational methods.

## Teaching Experience

**General Chemistry Laboratory I 2022 – 2023, 2026 – 2027**

Teaching Assistant, Temple University

- Led undergraduate laboratory sections, assisted with experiments, graded lab reports, and mentored students in foundational chemical concepts.

## Mentoring

- **Jay Ragvirsingh** (2024–2025), Undergraduate, Temple University.
- **Ozodabonu Zafarova** (2026 Summer), High school student, Temple University.

## Referees


**Dr. Spiridoula Matsika**
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